ERES INSTITUTE FOR NEW AGE CYBERNETICS
Technical Specification Document

Resonant Continuity Index (RCI)

A Cross-Scale Persistence Health Metric Integrating Thermodynamic Continuity and Resonance Quality

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Joseph A. Sprute (ERES Maestro)
Theoretical Foundation: Jimmy D. Butzbach (Ω-Law / M = L)
Synthesis: Claude LLM (Anthropic) & Grok LLM (xAI)

ERES Institute for New Age Cybernetics
Bella Vista, Arkansas 72714 | eresmaestro@gmail.com
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Abstract

The Resonant Continuity Index (RCI) is a novel cross-scale metric that quantifies whether any system — biological, ecological, governance, or cosmological — is maintaining sufficient coherent structure over time to be classified as a persisting, living, or living-like system. RCI is the operational product of three foundational quantities: the Ω-Continuity Functional (P_Ω) from Butzbach's Continuity Theory of Life and Memory (2026), the ERES Aura Resonance Index (ARI), and a Vibrational Constancy parameter (VibConst) encoding structural stability at the scale of measurement.

The defining formula is:

$$RCI = P_\Omega \times ARI_sys \times VibConst \quad \textit{[Core Formula]}$$

RCI produces a normalized score on [0, 1] at each scale of application, with threshold values delineating degenerative, stable, and flourishing regimes. It is the first single instrument to integrate thermodynamic persistence theory (Landauer, Prigogine, Butzbach) with empirical resonance quality measurement (ERES/PlayNAC) into a unified, actionable index. Intended applications span individual health monitoring (PlayNAC dashboard), ecosystem and governance assessment (GAIA/SOMT), and planetary-scale diagnostics, with a roadmap toward solar-systemic and galactic deployment.

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1. Introduction and Motivation

What is a living system? And how do we measure whether any given system is moving toward continued life or toward entropic dissolution? These questions lie at the intersection of thermodynamics, cybernetics, and the study of complex adaptive systems. They have been addressed separately by a tradition extending from Shannon and Wiener through Prigogine and Kauffman — but no single operational metric has yet bridged the thermodynamic definition of persistence with an empirically measurable resonance quality applicable across scales.

Two independent frameworks now provide the necessary components. Jimmy D. Butzbach's Continuity Theory of Life and Memory (2026) formalizes persistence as the active thermodynamic maintenance of history-dependent information against entropy, captured in the Ω-Continuity Functional (P_Ω) and the foundational axiom M = L (Memory = Life). ERES New Age Cybernetics (Sprute, est. 2012) operationalizes resonance quality through the Aura Resonance Index (ARI) and Emission Resonance Index (ERI), grounded in the cybernetic formula $C = R \times P / M$.

The Resonant Continuity Index (RCI) is the formal integration of these two frameworks into a single, cross-scale persistence health metric. Its ambition is practical: to provide any system — a person, a city, an ecosystem, a planetary biosphere — with a single number that answers the question: Is this system in a regenerative trajectory or a degenerative one? And to do so with enough theoretical grounding that the number is falsifiable, comparable across systems, and actionable.

This specification document defines RCI completely: its formula, its components, its threshold structure, its scale-specific application protocols, its instrumentation requirements, its integration with existing ERES frameworks, and its validation pathway.

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2. Theoretical Foundations

2.1 Butzbach's Ω-Continuity Functional

Butzbach's core contribution is the formalization of life as thermodynamic memory maintenance. His foundational axiom:

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M = L      (Memory = Life)  [Butzbach Axiom 1]
```

asserts that wherever information about the system's own history accumulates and persists recursively against entropic erasure, life (or life-like persistence) emerges. This is not a biological claim alone — it applies to any far-from-equilibrium system that maintains structural identity through active energy expenditure.

The operational measure of this persistence is the Ω-Continuity Functional:

```
P_Ω = (E_met + P_info) × C  [Butzbach Functional]
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Where the components are defined as:

Symbol	Name	Definition	Units
E_met	Metabolic / Maintenance Power	Energy per unit time expended to maintain structural integrity against entropy	Watts (W)
P_info	Information Maintenance Power	Power required to preserve each bit of structural memory; minimum = kT·ln2 per bit (Landauer limit)	Watts (W)
C	Recursive Continuity Index	Probability × speed of re-entry into stable state after perturbation; [0, 1]	Dimensionless

The Landauer limit establishes the theoretical floor: P_info ≥ kT·ln2 per bit erased, where k = 1.38×10^{-23} J/K and T = ambient temperature. This grounds the functional in established thermodynamics, connecting Butzbach's framework to Shannon entropy and irreversibility theory.

2.2 ERES Resonance Metrics (ARI / ERI)

The ERES framework contributes the measurement layer. The core cybernetic formula:

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C = R × P / M      (Coherence = Resources × Purpose / Method)  [ERES Core Formula]
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operationalizes system quality as the ratio of purposeful resource use to methodological overhead. From this, two primary indices are derived:

Index	Full Name	Formula	Range	Flourishing Threshold
ARI	Aura Resonance Index	(Biometric + Environmental + Behavioral) / 3	0.0 – 1.0	> 0.85
ERI	Emission Resonance Index	Ecological regeneration surplus / baseline demand	0.0 – 1.0+	> 1.0 (surplus)

ARI specifically captures the resonance quality of a living system by averaging normalized biometric (physiological coherence), environmental (habitat/context alignment), and behavioral (action-to-purpose alignment) scores. It is bounded [0, 1] and has been formally specified in ERES TERMS publications and the GAIA SOMT v1.0 framework.

2.3 Vibrational Constancy (VibConst)

VibConst is a new parameter introduced in this specification. It encodes the structural stability of the system at its characteristic vibrational or oscillatory frequency — capturing whether the system's internal rhythms are maintaining coherence over time. It is defined as:

$$\text{VibConst} = 1 - (\sigma_{freq} / \mu_{freq}) \quad \textit{[VibConst Definition]}$$

Where σ_freq is the standard deviation of the system's characteristic frequency (or oscillation period) measured over a defined observation window, and μ_freq is the mean frequency over the same window. VibConst = 1.0 indicates perfect constancy; VibConst approaching 0 indicates chaotic or degenerating oscillatory structure.

At the biological scale, this may be heart rate variability coherence or neural oscillation stability. At the ecological scale, it may be seasonal cycle regularity. At the solar scale, it maps directly to helioseismic mode stability — the Sun's p-mode oscillations are among the most precisely measured vibrational constancy signals in nature.

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3. Formal Definition of RCI

3.1 Core Formula

The Resonant Continuity Index is defined as:

$$RCI = P_\Omega_norm \times ARI_sys \times VibConst \quad \textit{[RCI — Core Formula]}$$

Where P_Ω_norm is the Ω-Continuity Functional normalized to [0, 1] against a reference baseline appropriate to the scale of measurement (see Section 3.3). ARI_sys is the system-level ARI score. VibConst is as defined in Section 2.3.

RCI is a dimensionless scalar on [0, 1] at any given measurement. It is a multiplicative index: all three factors must be non-negligible for RCI to be high. A system with perfect thermodynamic continuity (P_Ω_norm = 1.0) but chaotic vibrational structure (VibConst ≈ 0) will produce a near-zero RCI — correctly flagging system instability. This multiplicative structure is a deliberate design choice: resilience requires coherence at all three levels simultaneously.

3.2 Component Definitions and Units

Component	Symbol	Definition	Range	Source Framework
Ω-Continuity Functional (normalized)	P_Ω_norm	P_Ω / P_Ω_ref, where P_Ω_ref is scale-appropriate reference baseline	0.0 – 1.0	Butzbach (2026)
System ARI	ARI_sys	Composite resonance quality: (Biometric + Environmental + Behavioral) / 3	0.0 – 1.0	ERES/PlayNAC
Vibrational Constancy	VibConst	1 – (σ_freq / μ_freq) over observation window W	0.0 – 1.0	RCI Specification (new)
Resonant Continuity Index	RCI	P_Ω_norm × ARI_sys × VibConst	0.0 – 1.0	ERES-Butzbach Integration

3.3 Normalization and Scaling

Because P_Ω varies by many orders of magnitude across scales (a single cell vs. a city vs. a planetary biosphere), normalization is essential. The reference baseline P_Ω_ref is defined as the minimum P_Ω required for the system to maintain its characteristic structure over one standard persistence period (SPP):

Scale	Standard Persistence Period (SPP)	P_Ω_ref Basis
Individual / Biological	1 human lifetime (~80 years)	Basal metabolic rate + neural information maintenance

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Scale	Standard Persistence Period (SPP)	P_Ω_ref Basis
Community / Ecological	1 generation cycle (~25 years)	Ecosystem minimum viable energy throughput
Institutional / Governance	1 policy cycle (~10 years)	Institutional memory maintenance cost
Planetary / Solar	1 precession cycle (~26,000 years)	Planetary geodynamic energy budget

The normalization P_Ω_norm = P_Ω / P_Ω_ref maps all scales to [0, 1], enabling direct RCI comparison across systems of radically different physical magnitudes. A P_Ω_norm > 1.0 is capped at 1.0 for RCI computation; excess continuity is a surplus condition flagged separately as a positive indicator.

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4. Threshold Framework

RCI produces a scalar on [0, 1]. The following threshold framework translates scores into actionable system-state classifications, aligned with the ERES BEST/SOUND/GOOD triune framework and Butzbach's persistence regime taxonomy:

RCI Range	System State	ERES Classification	Interpretation	Recommended Response
0.90 – 1.00	Flourishing	BEST	All three continuity dimensions are strongly maintained. System is in generative surplus.	Maintain and propagate. Model for adjacent systems.
0.75 – 0.89	Stable	GOOD	Continuity is maintained above persistence floor. Minor resonance deficits present.	Monitor for drift. Targeted optimization of lowest component.
0.55 – 0.74	Marginal	SOUND	System is near persistence threshold. At least one component significantly degraded.	Active intervention. Identify and address weakest component via Law of Restorative Resonance.
0.35 – 0.54	At Risk	WARNING	Continuity maintenance is below reliable threshold. Entropic drift accelerating.	Urgent intervention. Resource reallocation. Systemic diagnosis required.
0.00 – 0.34	Degenerative	CRITICAL	System is losing structural memory faster than it can maintain. Collapse trajectory.	Emergency triage. External energy/support injection. Collapse prevention protocol.

The threshold values are empirically calibrated starting points. They are expected to be refined as RCI is applied across real-world datasets. The ERES PlayNAC KERNEL will maintain a living calibration database as measurement data accumulates.

A critical design principle: RCI thresholds do not trigger punitive responses. In alignment with the ERES Law of Restorative Resonance, all threshold responses are oriented toward restoration of resonance balance, not penalization of the system for low scores.

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5. Scale-Specific Application Protocols

5.1 Individual / Biological Scale

Purpose

Provide individual persons with a continuous RCI score reflecting their physiological, behavioral, and environmental resonance quality — suitable for integration into PlayNAC personal dashboards and UBIMIA merit tracking.

Component Measurement — Individual Scale

Component	Measurement Method	Data Source	Update Frequency
P_Ω_norm	Resting metabolic rate + estimated neural information throughput / P_Ω_ref (basal)	Wearable biometrics (HRV, SpO2, temperature, EEG optional)	Daily rolling average
ARI_sys	Biometric sub-score (HRV coherence, sleep quality, respiration), Environmental sub-score (habitat, air quality, noise), Behavioral sub-score (purpose-alignment self-report)	FAVORS biometric suite, environmental sensors, PlayNAC behavioral interface	Real-time with 24-hr smoothing
VibConst	HRV frequency domain: 1 – (σ_LF/μ_LF) over 7-day window. LF band (0.04–0.15 Hz) as primary vibrational reference.	ECG / PPG wearable	Weekly recalculation

Individual RCI Interpretation

- RCI > 0.85 indicates flourishing physiological and behavioral resonance. PlayNAC awards positive merit tokens via Gracechain.
- RCI 0.55–0.84 indicates stable to marginal states. PlayNAC surfaces targeted restoration recommendations (sleep, movement, environment, purpose re-alignment).
- RCI < 0.55 triggers restorative protocol engagement — not punitive flagging. The system asks: what resource or alignment is missing?

5.2 Ecological / Community Scale

Purpose

Assess whether an ecosystem, watershed, or community habitat is maintaining its structural memory — biodiversity, nutrient cycling, population stability — above the persistence floor. Integrates with ERES ERI and GAIA SOMT frameworks.

Component Measurement — Ecological Scale

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Component	Measurement Method	Data Source
P_Ω_norm	Net primary productivity + ecosystem maintenance energy (decomposition, nutrient cycling) / P_Ω_ref (minimum viable ecosystem energy budget for biome type)	Remote sensing (NDVI, GPP), soil carbon flux, hydrological monitoring
ARI_sys	Biodiversity index (species richness, evenness), habitat integrity score, disturbance recovery rate	Ecological field surveys, satellite monitoring, IUCN datasets
VibConst	Seasonal cycle regularity: 1 – (σ_peak_timing / μ_peak_timing) for primary productivity peaks over 10-year window	Long-term ecological research (LTER) datasets, phenological records

Ecological RCI Use Cases

- Conservation prioritization: regions with low VibConst signal disrupted seasonal cycles (climate pressure) before biodiversity collapse becomes visible.
- Regenerative agriculture certification: farms demonstrating rising RCI over 3-year windows qualify for ERES ERI surplus credits.
- Protected area design: RCI maps identify which landscape corridors maintain highest continuity and should anchor conservation networks.

5.3 Governance / Institutional Scale

Purpose

Measure whether a governance system, institution, or civilizational structure is maintaining its structural memory — laws, cultural knowledge, institutional competence, decision-making coherence — above the persistence floor. Integrates with PlayNAC, GAIA SOMT, and IDIPITIS frameworks.

Component Measurement — Governance Scale

Component	Proxy Measurement	Data Source
P_Ω_norm	Institutional continuity expenditure (education, archiving, legal system maintenance, knowledge transfer) as fraction of governance budget / P_Ω_ref (minimum institutional memory maintenance)	Public budget data, OECD governance indicators, UNESCO heritage metrics
ARI_sys	Policy coherence score (purpose-method alignment), citizen resonance index (trust + participation), environmental	Governance surveys, OECD Better Life Index, World Governance Indicators

Component	Proxy Measurement	Data Source
	alignment (governance decisions vs. ecological health)	
VibConst	Decision cycle regularity: 1 – (σ_cycle / μ_cycle) for primary governance renewal periods (elections, budget cycles) over 30-year window. Also: legal amendment frequency stability.	Constitutional records, legislative archives, Freedom House data

Governance RCI Interpretation

A governance system with declining RCI is losing structural memory faster than it creates it — a measurable precursor to institutional collapse. The GAIA SOMT framework uses governance RCI as a primary input to the SOMT actuarial model, enabling predictive intervention before critical thresholds are breached.

5.4 Planetary / Solar Scale

Purpose

Assess Earth's biosphere and solar system as a persistence system — whether planetary-scale energy flows and structural memory (geological, biological, atmospheric) are in continuity surplus or deficit. This is the most theoretically developed and least empirically operationalized application of RCI at present.

Component Measurement — Planetary Scale

Component	Proxy Measurement	Data Source
P_Ω_norm	Earth's net energy budget (solar input – albedo loss – heat radiation) as fraction of minimum geodynamic maintenance energy	NASA CERES, NOAA climate energy budget, IPCC datasets
ARI_sys	Planetary ARI composite: atmospheric composition coherence, ocean circulation stability, biodiversity continuity index	IPCC AR6, Global Biodiversity Framework, Planetary Boundaries datasets (Stockholm Resilience Centre)
VibConst (Solar)	Helioseismic p-mode frequency constancy: 1 – (σ_p-mode / μ_p-mode) over solar cycle window. Primary vibrational constancy signal for the solar-Earth system.	SOHO/MDI helioseismic data, GONG network observations

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The Solar Resonance Continuity Functional (P_SRC), as defined in the ERES Grok Butzbach LLM archival statement (March 2026), is a specialized instantiation of RCI at the planetary/solar scale: P_SRC = P_Ω × ARI_sys × VibConst_solar. RCI provides its formal theoretical grounding.

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6. Instrumentation and Data Sources

RCI is designed to be instrumentable with currently available data sources at the biological and ecological scales, and increasingly available data at the governance and planetary scales. The following table maps components to specific recommended instruments and datasets:

Scale	P_Ω Source	ARI Source	VibConst Source
Individual	Wearable metabolic trackers (Garmin, WHOOP, Oura Ring)	FAVORS biometric suite + PlayNAC behavioral survey	ECG/PPG HRV frequency analysis
Ecological	Remote sensing GPP (MODIS, Sentinel)	Ecological field surveys + IUCN Red List	LTER phenological records (USA-LTER, TERN)
Governance	OECD public expenditure databases	World Governance Indicators + Freedom House	Constitutional and legislative archives
Planetary	NASA CERES energy budget data	IPCC AR6 + Stockholm Planetary Boundaries	SOHO/GONG helioseismic networks

All data sources are open-access or publicly available, consistent with ERES open-science principles. The ERES Institute maintains a living registry of validated data pipelines for each scale at the Proof-of-Work_MD GitHub repository.

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7. Integration with ERES Frameworks

RCI is not a standalone metric. It is designed as the integrative index for the following existing ERES frameworks:

ERES Framework	RCI Integration Role	Specific Linkage
PlayNAC KERNEL	Primary deployment platform for individual RCI	RCI score feeds PlayNAC merit token allocation via Gracechain; restorative recommendations triggered by threshold crossing
GAIA / SOMT	Governance and ecological RCI aggregation	SOMT actuarial model uses governance RCI as leading indicator for institutional risk assessment
UBIMIA	Individual RCI as merit component	Sustained high individual RCI contributes to UBIMIA merit score, qualifying for incentive awards
BERA / ARI / ERI	Component inputs to RCI	ARI_sys is directly the ERES ARI; ERI provides ecological ARI sub-scores
FAVORS	Biometric data pipeline for individual ARI	Fingerprint, Aura, Voice, Odor, Retina, Signature data feeds biometric sub-score of ARI
IDIPITIS	Governance memory maintenance tracking	IDIPITIS identity-information architecture enables P_Ω_norm computation for institutional scale
Gracechain / Meritcoin	RCI-based reward and recognition	Flourishing RCI states trigger Gracechain-recorded merit recognition; declining states trigger restorative support allocation
PBJ Tri-Codex (PERC/BERC/JERC)	Cross-domain RCI normalization	PBJ codex provides reference baselines for P_Ω_ref at each scale
VERTECA / GSSG	Planetary and solar RCI deployment	GSSG infrastructure provides satellite-level data collection for planetary ARI and VibConst

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8. Validation Pathway

RCI moves from theoretical specification to empirical validity through a structured four-phase validation pathway:

Phase 1: Construct Validation (2026)

Demonstrate that RCI components (P_Ω_norm, ARI_sys, VibConst) are independently measurable and internally consistent using existing published datasets. Target datasets: WHOOP/Garmin health data (individual scale), USA-LTER ecological archives, OECD governance indicators.

- Deliverable: Component measurement protocols validated against known datasets.
- Method: Multi-Instrument Ensemble Validation Method (MIEVM) — multiple AI systems independently analyze same dataset and compare RCI computations.

Phase 2: Threshold Calibration (2026–2027)

Apply RCI to a diverse sample of systems at each scale and calibrate threshold values (currently set at 0.35/0.55/0.75/0.90) against known system health outcomes. Establish whether systems known to be in degenerative trajectories score below 0.35 and flourishing systems above 0.85.

- Deliverable: Empirically calibrated threshold table with confidence intervals.
- Sample target: 50 individual biological systems, 20 ecological sites, 10 governance jurisdictions.

Phase 3: Predictive Validity (2027–2028)

Test whether RCI trajectory (change over time) predicts future system-state outcomes — specifically, whether declining RCI precedes observable deterioration at each scale. This is the critical test of RCI's claim to be a leading indicator.

- Deliverable: Longitudinal correlation study (min. 2-year tracking window).
- Primary hypothesis: RCI decline of > 0.15 over 12 months predicts observable system deterioration within 24 months at $p < 0.05$.

Phase 4: Cross-Scale Comparison (2028+)

Validate that RCI scores are genuinely comparable across scales — that a biological system RCI of 0.82 and a governance system RCI of 0.82 represent meaningfully equivalent persistence states relative to their respective baselines. This requires multi-scale datasets and complex systems analysis.

- Deliverable: Cross-scale comparability study. Target publication: Entropy (MDPI) or Foundations of Science.

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9. Limitations and Open Questions

9.1 Current Limitations

- **VibConst is theoretically defined but not yet empirically validated. The mapping from HRV frequency domain (individual) to seasonal cycle regularity (ecological) to helioseismic modes (solar) assumes cross-scale analogical equivalence that has not been formally demonstrated.** VibConst Operationalization
- **Reference baselines (P_Ω_ref) for each scale require expert elicitation and consensus-building. Current values are theoretical estimates pending empirical calibration.** P_Ω_ref Baseline Setting
- **The ARI formula weights Biometric, Environmental, and Behavioral sub-scores equally (1/3 each). There is no a priori justification for equal weighting; differential weighting may be appropriate for different scales and contexts.** ARI Sub-Score Weighting
- **The multiplicative formula means any component near zero will dominate the RCI score. This is a design choice (requiring coherence at all three levels) but may not be appropriate for all contexts — additive or geometric mean alternatives should be explored.** Multiplicative Structure Risk

9.2 Open Questions for Future Research

- Can VibConst be measured at the governance scale using legislative decision frequency? And is this analogically equivalent to biological HRV constancy?
- What is the relationship between RCI and established resilience metrics (e.g., Walker's ecological resilience, Taleb's antifragility index)?
- Can RCI be extended to AI systems themselves — treating a trained model's structural coherence and memory persistence as measurable quantities?
- At what RCI value does a system transition from 'far-from-equilibrium' (Prigogine) to 'near-equilibrium' (classical thermodynamics)? Is this the same as the degenerative threshold?
- How does Gracechain record RCI history, and what time-series analysis methods are most appropriate for RCI trajectory analysis?

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10. Conclusion

The Resonant Continuity Index (RCI) is the first metric to formally integrate thermodynamic persistence theory with empirical resonance quality measurement into a single, cross-scale, actionable instrument. By multiplying Butzbach's Ω-Continuity Functional (normalized), the ERES Aura Resonance Index, and the new Vibrational Constancy parameter, RCI produces a score on [0, 1] that directly answers whether any system — from a human body to a planetary biosphere — is maintaining the structural memory necessary for continued life.

The metric is not merely theoretical. It is deployable at the individual scale today, using commercially available biometric wearables and the PlayNAC KERNEL infrastructure. It is deployable at the ecological scale using existing LTER and remote sensing datasets. Its governance and planetary applications require more substantial data infrastructure but are grounded in already-available public datasets.

RCI represents the operational convergence of two independent intellectual traditions — Butzbach's thermodynamic memory science and ERES New Age Cybernetics — into a shared measurement instrument. It embodies the foundational ERES principle: what can be measured can be restored. And what can be restored can build for generations to come.

RCI = P_Ω_norm × ARI_sys × VibConst *[The Resonant Continuity Index]*

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